DC Charging (CCS2)-IV

Introduction

Imagine pulling up to a public charging station, plugging in your electric vehicle (EV), and expecting a seamless, efficient power transfer. What happens behind the scenes to make this possible? Beyond the physical connection, a complex digital conversation unfolds between your EV and the charging station (EVSE). This lecture delves into the intricate world of high-level communication (HLC) in DC charging, specifically focusing on the Combined Charging System (CCS2) standard. We explore how the foundational principles of data networking, embodied by the 7-layer Open Systems Interconnection (OSI) model, are adapted to ensure reliable and intelligent power delivery. Understanding these layers is crucial for appreciating the robustness and interoperability of modern EV charging infrastructure, ensuring that your vehicle can communicate effectively with any compliant charger, regardless of its manufacturer or location.

The core of this digital dialogue relies on Power Line Communication (PLC), which transmits data over the same control pilot (CP) and protective earth (PE) lines used for signaling. Before any significant data exchange can occur, a critical procedure known as SLAC (Signal Level Attenuation Characterization) takes place. This process ensures that the correct PLC modems—those physically connected—are logically paired, preventing miscommunication in environments with multiple chargers and EVs. This lecture will break down how each layer of the OSI model contributes to this sophisticated communication, from establishing a stable physical link to managing application-level charging parameters, providing a comprehensive view of the digital architecture underpinning CCS2 DC charging.

Core Concepts

Term	Definition	Significance
`CCS2 HLC`	Combined Charging System 2 High-Level Communication, the digital dialogue between an EV and EVSE for DC charging.	Enables intelligent control and monitoring of the charging process, exchanging critical parameters like voltage, current, and battery status.
`OSI Model`	Open Systems Interconnection model, a conceptual framework that standardizes the functions	Provides a universal structure for understanding and designing network

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	of a telecommunication or computing system into seven layers.	communication, ensuring interoperability between diverse systems like EVs and EVSEs.
SLAC Procedure	Signal Level Attenuation Characterization, a process to logically connect the PLC modems of the EV and EVSE.	Crucial for identifying and pairing the physically connected EV and EVSE modems, especially in multi-charger environments, by measuring signal attenuation.
PLC Modems	Power Line Communication modems, devices that enable data transmission over existing power lines (CP and PE in CCS2).	Facilitate the high-level communication in CCS2 by using the same physical wires as the power signaling, reducing complexity and wiring.
Physical Layer	The lowest layer (Layer 1) of the OSI model, responsible for the physical connection and transmission of raw bit streams over a physical medium.	Establishes the fundamental electrical and mechanical interface, defining voltage levels, frequency bands, and physical link stability for data transfer.
Data Link Layer	The second layer (Layer 2) of the OSI model, responsible for node-to-node data transfer and error detection/correction within a local network segment.	Manages physical addressing (MAC addresses), performs the SLAC procedure, and ensures reliable, error-free data frame transfer between directly connected devices.
Network Layer	The third layer (Layer 3) of the OSI model, responsible for logical addressing, routing, and path determination across different networks.	Assigns unique IP addresses to devices, enabling data packets to be routed efficiently across potentially complex network infrastructures to their correct destination.

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`IP Address`	Internet Protocol address, a numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.	Provides a logical address for devices, allowing them to be uniquely identified and located within a network for routing data.
`MAC Address`	Media Access Control address, a unique physical hardware identifier assigned to network interfaces for communications within a network segment.	Serves as a permanent, unique physical address for network interface cards, crucial for direct device-to-device communication within a local network.
`ISO 15118`	An international standard defining the communication interface between an EV and EVSE for charging.	Specifies the protocols and requirements for high-level communication in CCS2, ensuring interoperability and advanced charging functionalities.

Detailed Analysis

1. CCS2 High-Level Communication (HLC) and the OSI Model

High-level communication (HLC) in CCS2 is essential for exchanging various charging parameters and information between the EV and EVSE. This includes data such as voltage, current, State of Charge (SOC), temperature, State of Health (SOH), and charging time. This communication occurs over Power Line Communication (PLC) on the Control Pilot (CP) and Protective Earth (PE) lines, leveraging the existing physical connection. The entire process of message exchange adheres to the 7-layer Open Systems Interconnection (OSI) model, a standardized framework for data transfer between networked computers. Each message traverses these layers, starting from the physical connection and progressing up to the application level.

The OSI model, standardized by ISO in 1984, breaks down network communication into seven distinct layers, each with specific protocols and functions. These layers ensure successful communication even between systems with different architectures or across diverse networks. The layers, from top to bottom, are:
* **Application Layer (Layer 7)**: Provides network services to end-user applications.
* **Presentation Layer (Layer 6)**: Translates data between the application layer and

the network format, handling encryption/decryption and compression.
* **Session Layer (Layer 5)**: Manages communication sessions between applications, including authentication and authorization.
* **Transport Layer (Layer 4)**: Ensures reliable end-to-end data delivery, including segmentation, flow control, and error control.
* **Network Layer (Layer 3)**: Handles logical addressing and routing of data packets across networks.
* **Data Link Layer (Layer 2)**: Manages physical addressing (MAC addresses) and error-free data transfer within a local network segment.
* **Physical Layer (Layer 1)**: Defines the electrical and mechanical specifications for the physical connection and raw bit transmission.

In the context of CCS2, these layers are defined by the ISO 15118 standard. ISO 15118-1 provides general information and requirements. ISO 15118-3 specifically defines the protocols for the Physical and Data Link layers (Layer 1 and Layer 2). The higher layers (Layer 3 to Layer 7) are detailed in ISO 15118-2, which derives its protocols from established networking standards like Internet Protocol (IP) for the Network Layer, User Datagram Protocol (UDP) or Transmission Control Protocol (TCP) for the Transport Layer, XML/EXI for the Presentation Layer, and V2G application layer protocols for the Application Layer.

2. OSI Layer 1: The Physical Foundation

The Physical Layer (Layer 1) is the bedrock of communication in CCS2 HLC. Its primary role is to establish and stabilize the physical link between the EV and EVSE.

- **Initial Condition**: Before Layer 1 protocols are implemented, the protective earth (PE) and control pilot (CP) lines must be physically connected, indicating that the charger plug is inserted. The charging stage is typically in a "B2 state," characterized by a Pulse Width Modulation (PWM) signal of 3% to 7% on the CP line. Both the EV and EVSE are equipped with compatible HomePlug Green PHY-based PLC modems.
- **Function**: The protocols at this layer work to stabilize the physical link on the PE-CP wires, ensuring a robust connection to the opposing side (EV to EVSE, or EVSE to EV). This involves defining the physical properties of the communication medium, such as voltage levels and the frequency band for data transmission.
- **Outcome**: A stable physical link is established, and the PLC modems on both the EV and EVSE are ready for communication within a defined frequency band, typically 2 MHz to 28 MHz. This layer ensures bidirectional data transfer, meaning information can flow from EV to EVSE and vice-versa.

3. OSI Layer 2: Data Link and Logical Connection

Building upon the stable physical link, the Data Link Layer (Layer 2) focuses on establishing a logical connection and ensuring reliable data transfer within the local segment.

- **Initial Condition**: The Physical Layer protocols must have been successfully implemented, meaning a stable physical link is already in place.
- **Function**: This layer is responsible for configuring the PLC nodes (EV and EVSE). A crucial procedure performed here is the Signal Level Attenuation Characterization (SLAC). SLAC measures the signal attenuation between PLC modems to identify the physically connected pair, especially important in charging stations with multiple EVSEs. This ensures that the EV's PLC modem communicates only with its physically connected EVSE's modem, preventing miscommunication. Once SLAC is complete, a logical connection is set up between the identified PLC modems. This layer also handles physical addressing using MAC (Media Access Control) addresses. Each network interface card (NIC) in a device (like an EV or EVSE) has a unique, 12-digit alphanumeric MAC address embedded in its hardware, which remains constant.
- **Outcome**: A stable communication link is established through the logical connection of the PLC modems. This layer ensures an error-free transfer of data frames from one node to another over the physical media. The IP packets received from the Network Layer are augmented with the MAC addresses of the sender and receiver, along with error control codes, to form data frames.

4. OSI Layer 3: Network Addressing and Routing

The Network Layer (Layer 3) takes the logical connection a step further by assigning unique addresses and determining the optimal path for data across potentially diverse networks.

- **Initial Condition**: The Data Link Layer protocols must have been successfully implemented, meaning a logical connection between the PLC modems is established.
- **Function**: This layer implements the Internet Protocol version 6 (IPv6) to assign unique logical addresses (IP addresses) to all communicating nodes, including the EV's PLC modem, the EVSE's PLC modem, and other communication controllers within both systems. The IP address is dynamic and can change depending on the network the device is connected to. The Stateless Address Autoconfiguration (SLAAC) procedure is used to assign

these IP addresses. Additionally, this layer implements ICMPv6 (Internet Control Message Protocol version 6) for sending error messages. A key function is routing, where the layer determines the best path for data to travel from the source to the destination, even if it involves multiple intermediate networks.

- **Outcome**: All nodes involved in the communication (EV, EVSE, and their respective communication controllers) obtain valid and unique IP addresses. This logical addressing enables efficient routing of data packets, ensuring they reach the correct destination across the network infrastructure. Data segments from the Transport Layer are augmented with sender and receiver IP addresses to form IP packets.

Key Takeaways

- [ ] High-level communication (HLC) in CCS2 DC charging uses the 7-layer OSI model for structured data exchange.
- [ ] Power Line Communication (PLC) over the Control Pilot (CP) and Protective Earth (PE) lines is the physical medium for HLC.
- [ ] The SLAC (Signal Level Attenuation Characterization) procedure is critical for logically connecting the correct EV and EVSE PLC modems.
- [ ] The Physical Layer (Layer 1) establishes the stable electrical link and defines properties like voltage levels and frequency bands (2-28 MHz).
- [ ] The Data Link Layer (Layer 2) manages physical addressing (MAC addresses) and ensures error-free data frame transfer, incorporating SLAC.
- [ ] The Network Layer (Layer 3) assigns unique logical IP addresses using IPv6 and SLAAC, and handles routing and path determination for data packets.
- [ ] ISO 15118 standards (specifically 15118-1, 15118-2, and 15118-3) define the protocols for each OSI layer in CCS2 HLC.
- [ ] MAC addresses are unique, permanent hardware identifiers, while IP addresses are dynamic logical identifiers for network routing.

Conclusion

The detailed exploration of the OSI model in the context of CCS2 DC charging reveals the sophisticated engineering required for seamless electric vehicle charging. From the moment an EV is plugged in, a meticulously orchestrated sequence of communication protocols, spanning all seven layers, ensures that power is delivered safely, efficiently, and intelligently. The physical layer establishes the fundamental electrical connection, while the data link layer, through procedures

like SLAC, guarantees that the right devices are talking to each other. The network layer then provides the logical addressing and routing necessary for data to traverse complex networks, ensuring that charging parameters and vehicle status updates reach their intended destinations.

This layered approach, as defined by the ISO 15118 standards, is not merely an academic exercise; it is the backbone of interoperability in the EV ecosystem. It allows EVs from different manufacturers to communicate with various charging stations, fostering a robust and accessible charging infrastructure. As EV adoption accelerates, the reliability and intelligence of this high-level communication will become even more critical, enabling advanced features like smart charging, vehicle-to-grid (V2G) capabilities, and optimized energy management. Understanding these underlying communication principles is fundamental for anyone involved in the design, deployment, or use of modern EV charging solutions, highlighting how digital intelligence is as vital as electrical power in the future of transportation.